

- **Database/schema**

Figure A1. Creation and Selection of the SupermarketDB Database

Shows the creation of the SupermarketDB database and the selection of the database for subsequent SQL operations.

```
MariaDB [(none)]> CREATE DATABASE SupermarketDB;  
Query OK, 1 row affected (0.004 sec)  
  
MariaDB [(none)]> USE SupermarketDB;  
Database changed
```

- **Tables**

Figure A2. SupermarketDB Table Listing

Shows the tables implemented in the SupermarketDB database, including Customer, Employee, Product, Inventory, Purchase, Sale, and their related tables and views

```
MariaDB [SupermarketDB]> SHOW TABLES;  
+-----+  
| Tables_in_supermarketdb |  
+-----+  
| activediscounts  
| category  
| customer  
| customerpurchasehistory  
| discount  
| employee  
| inventory  
| inventorymovement  
| loyaltycard  
| product  
| productdiscount  
| productspercategory  
| productstockstatus  
| purchase  
| purchaseitem  
| sale  
| saleitem  
| salessummary  
| supplier  
| supplierpurchasesummary  
+-----+  
20 rows in set (0.067 sec)
```

- **Table structures: showing primary keys, foreign keys**

Figures A3. Product & Sale Table Structure

Shows the structure of the Product & Sale table, including its attributes, data types, primary key, foreign keys, default values, and unique constraints.

Sample:

```
MariaDB [SupermarketDB]> DESCRIBE Product;
```

Field	Type	Null	Key	Default	Extra
productID	varchar(10)	NO	PRI	NULL	
categoryID	varchar(10)	NO	MUL	NULL	
supplierID	varchar(10)	NO	MUL	NULL	
productName	varchar(100)	NO	UNI	NULL	
unitPrice	decimal(10,2)	NO		NULL	
costPrice	decimal(10,2)	NO		NULL	
expiryDate	date	YES		NULL	
status	enum('Available','Out of Stock','Discontinued')	YES		Available	
barcode	varchar(50)	YES	UNI	NULL	


```
MariaDB [SupermarketDB]> DESCRIBE Sale;
```

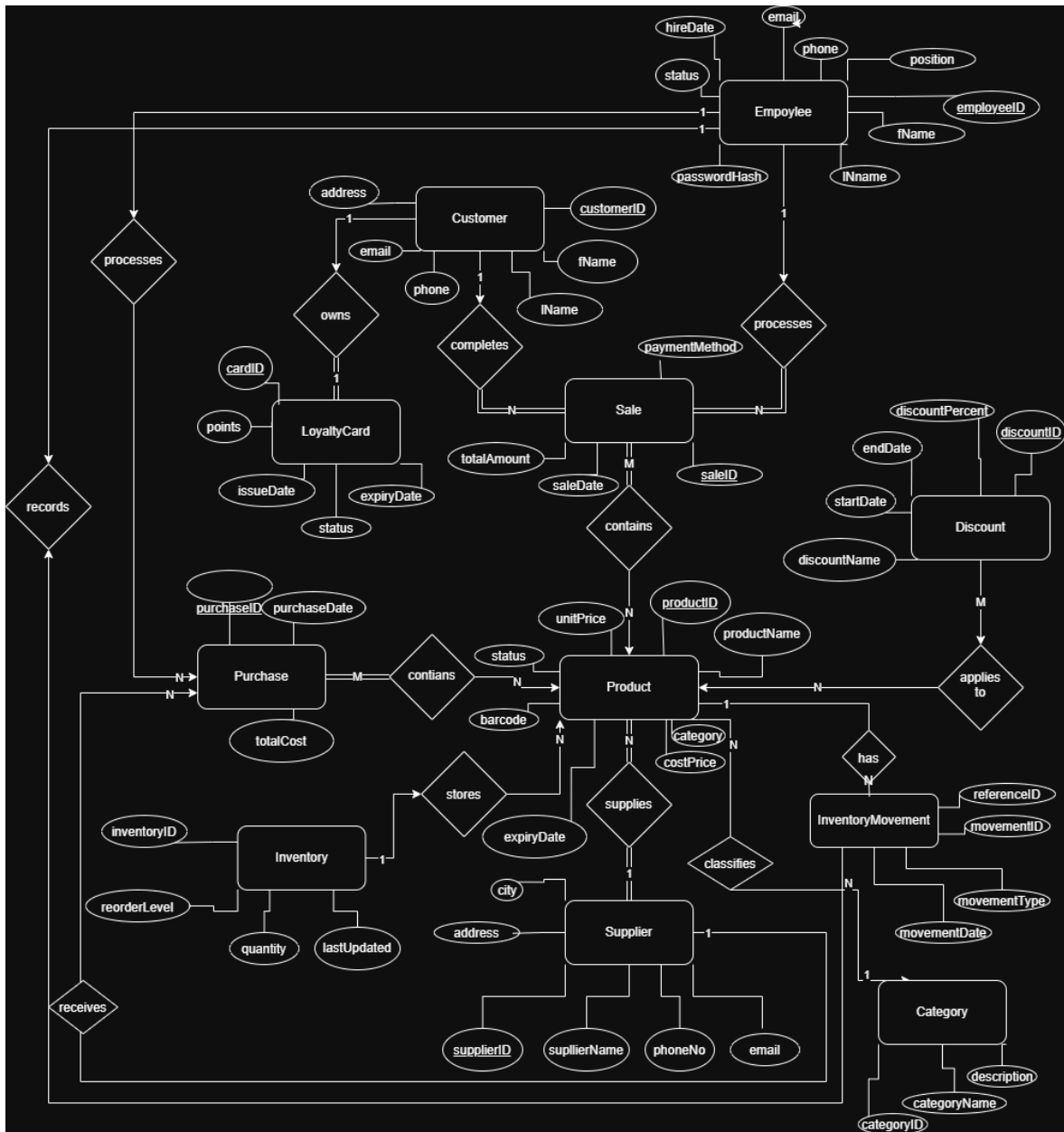
Field	Type	Null	Key	Default	Extra
saleID	varchar(10)	NO	PRI	NULL	
customerID	varchar(10)	YES	MUL	NULL	
employeeID	varchar(10)	NO	MUL	NULL	
saleDate	datetime	YES		current_timestamp()	
totalAmount	decimal(12,2)	YES		0.00	
paymentMethod	enum('Cash','Card','Mobile Money')	NO		NULL	
taxAmount	decimal(12,2)	NO		0.00	
pointsRedeemed	int(11)	NO		0	

8 rows in set (0.061 sec)

• Relationships

Figure A4. SupermarketDB Entity-Relationship Diagram

Shows the entities, attributes, primary and foreign key relationships, and cardinalities implemented in the SupermarketDB database.



- **Sample records**

Figure A5. Sample Supplier Records

Shows sample supplier records inserted into the SupermarketDB database.

```
MySQL Client (MariaDB 12.2 (x64)) - "C:\Program Files\MariaDB 12.2\bin\mysql.exe" --defaults-file=C:\Program Files\MariaDB 12.2\data\my.ini -u root -p

MariaDB [SupermarketDB]> INSERT INTO Supplier (supplierID, supplierName, phoneNo, email, address, city) VALUES
-> ('SUP001', 'Accra Foods Ltd', '+233240000001', 'info@accrafoods.com', '12 Ring Road', 'Accra'),
-> ('SUP002', 'Golden Harvest Ltd', '+233240000002', 'sales@goldenharvest.com', '25 Spintex Road', 'Accra'),
-> ('SUP003', 'Fresh Choice Ghana', '+233240000003', 'info@freshchoice.com', '8 Liberation Road', 'Accra'),
-> ('SUP004', 'Kumasi Wholesale', '+233240000004', 'sales@kumasiwholesale.com', '15 Adum Road', 'Kumasi'),
-> ('SUP005', 'Prime Beverages', '+233240000005', 'info@primebeverages.com', '22 Industrial Area', 'Tema'),
-> ('SUP006', 'Healthy Dairy Ltd', '+233240000006', 'sales@healthydairy.com', '17 Lake Road', 'Accra'),
-> ('SUP007', 'Best Bakery Supply', '+233240000007', 'info@bestbakery.com', '9 Market Street', 'Kumasi'),
-> ('SUP008', 'Royal Canned Foods', '+233240000008', 'info@royalcanned.com', '11 Factory Road', 'Accra'),
-> ('SUP009', 'Snack World Ltd', '+233240000009', 'sales@snackworld.com', '18 North Industrial Area', 'Accra'),
-> ('SUP010', 'Morning Foods Ltd', '+233240000010', 'info@morningfoods.com', '7 Airport Road', 'Accra'),
-> ('SUP011', 'Golden Grain Ltd', '+233240000011', 'sales@goldengrain.com', '14 Industrial Road', 'Tema'),
-> ('SUP012', 'Pure Oils Ghana', '+233240000012', 'sales@pureoils.com', '10 Oil Road', 'Tema'),
-> ('SUP013', 'Clean Home Ltd', '+233240000013', 'info@cleanhome.com', '20 Household Road', 'Accra'),
-> ('SUP014', 'Stationery Hub Ltd', '+233240000014', 'sales@stationeryhub.com', '13 School Road', 'Accra'),
-> ('SUP015', 'Sweet Treats Ltd', '+233240000015', 'sales@sweettreats.com', '2 Chocolate Road', 'Accra'),
-> ('SUP016', 'Pure Water Ghana', '+233240000016', 'info@purewaterghana.com', '30 Water Road', 'Tema');
Query OK, 16 rows affected (0.029 sec)
Records: 16 Duplicates: 0 Warnings: 0

MariaDB [SupermarketDB]> INSERT INTO Customer (customerID, fName, lName, email, address, phone) VALUES
-> ('CUS001', 'John', 'Mensah', 'john.mensah@gmail.com', 'East Legon', '+233500000001'),
-> ('CUS002', 'Ama', 'Boateng', 'ama.boateng@gmail.com', 'Osu', '+233500000002'),
-> ('CUS003', 'Kwame', 'Asante', 'kwame.asante@gmail.com', 'Kumasi', '+233500000003'),
-> ('CUS004', 'Akosua', 'Owusu', 'akosua.owusu@gmail.com', 'Madina', '+233500000004'),
-> ('CUS005', 'Daniel', 'Kofi', 'daniel.kofi@gmail.com', 'Tema', '+233500000005'),
-> ('CUS006', 'Abena', 'Osei', 'abena.osei@gmail.com', 'Adenta', '+233500000006'),
-> ('CUS007', 'Michael', 'Addo', 'michael.addo@gmail.com', 'Spintex', '+233500000007'),
-> ('CUS008', 'Linda', 'Amoah', 'linda.amoah@gmail.com', 'Dansoman', '+233500000008'),
-> ('CUS009', 'Emmanuel', 'Darko', 'emmanuel.darko@gmail.com', 'Kaneshie', '+233500000009'),
-> ('CUS010', 'Grace', 'Yeboah', 'grace.yeboah@gmail.com', 'Airport', '+233500000010'),
-> ('CUS011', 'Samuel', 'Appiah', 'samuel.appiah@gmail.com', 'Achimota', '+233500000011'),
-> ('CUS012', 'Mary', 'Frimpong', 'mary.frimpong@gmail.com', 'Lapaz', '+233500000012'),
-> ('CUS013', 'David', 'Antwi', 'david.antwi@gmail.com', 'Teshie', '+233500000013'),
-> ('CUS014', 'Esther', 'Badu', 'esther.badu@gmail.com', 'Nungua', '+233500000014'),
-> ('CUS015', 'Joseph', 'Adjei', 'joseph.adjei@gmail.com', 'Labadi', '+233500000015'),
-> ('CUS016', 'Patricia', 'Agyeman', 'patricia.agyeman@gmail.com', 'Cantonments', '+233500000016'),
```

Figure A6. Sample Customer Records * 10 Sale Records

Shows sample customer records inserted into the SupermarketDB database and displays 10 sale records

```
MySQL Client (MariaDB 12.2 (x64)) - "C:\Program Files\MariaDB 12.2\bin\mysql.exe" --defaults-file=C:\Program Files\MariaDB 12.2\data\my.ini -u root -p

MariaDB [SupermarketDB]> INSERT INTO Supplier (supplierID, supplierName, phoneNo, email, address, city) VALUES
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-> ('SUP003', 'Fresh Choice Ghana', '+233240000003', 'info@freshchoice.com', '8 Liberation Road', 'Accra'),
-> ('SUP004', 'Kumasi Wholesale', '+233240000004', 'sales@kumasiwholesale.com', '15 Adum Road', 'Kumasi'),
-> ('SUP005', 'Prime Beverages', '+233240000005', 'info@primebeverages.com', '22 Industrial Area', 'Tema'),
-> ('SUP006', 'Healthy Dairy Ltd', '+233240000006', 'sales@healthydairy.com', '17 Lake Road', 'Accra'),
-> ('SUP007', 'Best Bakery Supply', '+233240000007', 'info@bestbakery.com', '9 Market Street', 'Kumasi'),
-> ('SUP008', 'Royal Canned Foods', '+233240000008', 'info@royalcanned.com', '11 Factory Road', 'Accra'),
-> ('SUP009', 'Snack World Ltd', '+233240000009', 'sales@snackworld.com', '18 North Industrial Area', 'Accra'),
-> ('SUP010', 'Morning Foods Ltd', '+233240000010', 'info@morningfoods.com', '7 Airport Road', 'Accra'),
-> ('SUP011', 'Golden Grain Ltd', '+233240000011', 'sales@goldengrain.com', '14 Industrial Road', 'Tema'),
-> ('SUP012', 'Pure Oils Ghana', '+233240000012', 'sales@pureoils.com', '10 Oil Road', 'Tema'),
-> ('SUP013', 'Clean Home Ltd', '+233240000013', 'info@cleanhome.com', '20 Household Road', 'Accra'),
-> ('SUP014', 'Stationery Hub Ltd', '+233240000014', 'sales@stationeryhub.com', '13 School Road', 'Accra'),
-> ('SUP015', 'Sweet Treats Ltd', '+233240000015', 'sales@sweettreats.com', '2 Chocolate Road', 'Accra'),
-> ('SUP016', 'Pure Water Ghana', '+233240000016', 'info@purewaterghana.com', '30 Water Road', 'Tema');
Query OK, 16 rows affected (0.029 sec)
Records: 16 Duplicates: 0 Warnings: 0

MariaDB [SupermarketDB]> INSERT INTO Customer (customerID, fName, lName, email, address, phone) VALUES
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-> ('CUS002', 'Ama', 'Boateng', 'ama.boateng@gmail.com', 'Osu', '+233500000002'),
-> ('CUS003', 'Kwame', 'Asante', 'kwame.asante@gmail.com', 'Kumasi', '+233500000003'),
-> ('CUS004', 'Akosua', 'Owusu', 'akosua.owusu@gmail.com', 'Madina', '+233500000004'),
-> ('CUS005', 'Daniel', 'Kofi', 'daniel.kofi@gmail.com', 'Tema', '+233500000005'),
-> ('CUS006', 'Abena', 'Osei', 'abena.osei@gmail.com', 'Adenta', '+233500000006'),
-> ('CUS007', 'Michael', 'Addo', 'michael.addo@gmail.com', 'Spintex', '+233500000007'),
-> ('CUS008', 'Linda', 'Amoah', 'linda.amoah@gmail.com', 'Dansoman', '+233500000008'),
-> ('CUS009', 'Emmanuel', 'Darko', 'emmanuel.darko@gmail.com', 'Kaneshie', '+233500000009'),
-> ('CUS010', 'Grace', 'Yeboah', 'grace.yeboah@gmail.com', 'Airport', '+233500000010'),
-> ('CUS011', 'Samuel', 'Appiah', 'samuel.appiah@gmail.com', 'Achimota', '+233500000011'),
-> ('CUS012', 'Mary', 'Frimpong', 'mary.frimpong@gmail.com', 'Lapaz', '+233500000012'),
-> ('CUS013', 'David', 'Antwi', 'david.antwi@gmail.com', 'Teshie', '+233500000013'),
-> ('CUS014', 'Esther', 'Badu', 'esther.badu@gmail.com', 'Nungua', '+233500000014'),
-> ('CUS015', 'Joseph', 'Adjei', 'joseph.adjei@gmail.com', 'Labadi', '+233500000015'),
-> ('CUS016', 'Patricia', 'Agyeman', 'patricia.agyeman@gmail.com', 'Cantonments', '+233500000016'),
```

```
MariaDB [SupermarketDB]> SELECT * FROM Sale LIMIT 10;
```

saleID	customerID	employeeID	saleDate	totalAmount	paymentMethod	taxAmount	pointsRedeemed
SAL001	CUS001	EMP001	2026-08-01 00:00:00	528.65	Cash	15.40	10
SAL002	CUS002	EMP004	2026-08-01 00:00:00	23.48	Card	0.68	0
SAL003	CUS003	EMP007	2026-08-01 00:00:00	122.36	Mobile Money	3.56	0
SAL004	CUS004	EMP002	2026-08-01 00:00:00	282.22	Cash	8.22	20
SAL005	CUS005	EMP005	2026-08-02 00:00:00	49.90	Card	1.45	0
SAL006	CUS006	EMP008	2026-08-02 00:00:00	254.72	Mobile Money	7.42	0
SAL007	CUS007	EMP003	2026-08-02 00:00:00	280.16	Cash	8.16	0
SAL008	CUS008	EMP006	2026-08-02 00:00:00	18.39	Card	0.54	0
SAL009	CUS009	EMP001	2026-08-03 00:00:00	181.28	Mobile Money	5.28	10
SAL010	CUS010	EMP004	2026-08-03 00:00:00	148.32	Cash	4.32	10

10 rows in set (0.054 sec)

• Important SQL queries & results

Advanced queries:

Figure A7. Employee Sales Performance Query and Results

Shows an SQL query that calculates the number of transactions and total revenue processed by each employee, filtering employees based on total revenue and ordering the results by revenue.

```
MySQL Client (MariaDB 12.2 (x64)) - "C:\Program Files\MariaDB 12.2\bin\mysql.exe" --defaults-file="C:\Program Files\MariaDB 12.2\data\my.ini" --uroot -p
```

56 rows in set (0.003 sec)

```
MariaDB [SupermarketDB]> SELECT
->   e.employeeID,
->   CONCAT(e.fName, ' ', e.lName) AS cashierName,
->   e.position,
->   COUNT(s.saleID) AS totalTransactions,
->   SUM(s.totalAmount) AS totalRevenueProcessed
-> FROM Employee e
-> JOIN Sale s
->   ON e.employeeID = s.employeeID
-> GROUP BY
->   e.employeeID,
->   e.fName,
->   e.lName,
->   e.position
-> HAVING totalRevenueProcessed >= 100.00
-> ORDER BY totalRevenueProcessed DESC;
```

employeeID	cashierName	position	totalTransactions	totalRevenueProcessed
EMP002	Sarah Owusu	Cashier	4	997.86
EMP001	James Mensah	Manager	4	970.31
EMP004	Linda Adjai	Supervisor	4	827.14
EMP007	David Osei	Accountant	4	707.55
EMP008	Mary Boateng	Cashier	4	683.71
EMP003	Peter Asante	Cashier	3	490.59
EMP005	Michael Kofi	Stock Clerk	4	347.78
EMP006	Grace Amoah	Cashier	3	325.33

8 rows in set (0.027 sec)

```
MariaDB [SupermarketDB]> SELECT
->   cat.categoryID,
->   cat.categoryName,
->   SUM(si.quantity) AS totalUnitsSold,
->   SUM(si.quantity * si.unitPrice) AS grossRevenue,
```

Figure A8. Customer Purchase and Loyalty Analysis Query and Results

Shows an SQL query that combines customer, loyalty card, and sales data to calculate loyalty points, total orders, and total spending for customers.

```

MySQL Client (MariaDB 12.2 (x64)) - "C:\Program Files\MariaDB 12.2\bin\mysql.exe" --defaults-file=C:\Program Files\MariaDB 12.2\data\my.ini -uroot -p
MariaDB [SupermarketDB]> SELECT
-> c.customerID,
-> CONCAT(c.fName, ' ', c.lName) AS customerName,
-> c.phone,
-> lc.points AS loyaltyPoints,
-> COUNT(s.saleID) AS totalOrders,
-> SUM(s.totalAmount) AS totalSpent
-> FROM Customer c
-> JOIN LoyaltyCard lc
-> ON c.customerID = lc.customerID
-> JOIN Sale s
-> ON c.customerID = s.customerID
-> GROUP BY
-> c.customerID,
-> c.fName,
-> c.lName,
-> c.phone,
-> lc.points
-> ORDER BY totalSpent DESC
-> LIMIT 5;

```

customerID	customerName	phone	loyaltyPoints	totalOrders	totalSpent
CUS001	John Mensah	+233500000001	90	2	924.43
CUS004	Akosua Owusu	+233500000004	145	2	475.04
CUS002	Ama Boateng	+233500000002	180	2	429.30
CUS003	Kwame Asante	+233500000003	300	2	404.37
CUS020	Sarah Aidoo	+233500000020	300	1	289.43

5 rows in set (0.046 sec)

Figure A9. SQL Statements for Creating Database Views

Shows the SQL statements used to create the database views, including CustomerPurchaseHistory, ActiveDiscounts, SupplierPurchaseSummary, and SalesSummary.

```

MySQL Client (MariaDB 12.2 (x64)) - "C:\Program Files\MariaDB 12.2\bin\mysql.exe" --defaults-file=C:\Program Files\MariaDB 12.2\data\my.ini -uroot -p
MariaDB [SupermarketDB]> CREATE VIEW CustomerPurchaseHistory AS
-> SELECT c.customerID, CONCAT(c.fName, ' ', c.lName) AS customerName, s.saleID, s.saleDate,
-> p.productName, si.quantity, si.unitPrice, si.discount, si.subTotal
-> FROM Customer c
-> JOIN Sale s ON c.customerID = s.customerID
-> JOIN SaleItem si ON s.saleID = si.saleID
-> JOIN Product p ON si.productID = p.productID;
Query OK, 0 rows affected (0.014 sec)

MariaDB [SupermarketDB]> CREATE VIEW ActiveDiscounts AS
-> SELECT d.discountID, d.discountName, d.discountType, d.discountPercent, d.fixedAmount,
-> d.startDate, d.endDate, p.productID, p.productName
-> FROM Discount d
-> JOIN ProductDiscount pd ON d.discountID = pd.discountID
-> JOIN Product p ON pd.productID = p.productID
-> WHERE CURDATE() BETWEEN d.startDate AND d.endDate;
Query OK, 0 rows affected (0.042 sec)

MariaDB [SupermarketDB]>
MariaDB [SupermarketDB]> CREATE VIEW SupplierPurchaseSummary AS
-> SELECT s.supplierID, s.supplierName, COUNT(DISTINCT pu.purchaseID) AS totalPurchases,
-> COALESCE(SUM(pu.totalCost),0) AS totalPurchaseCost
-> FROM Supplier s
-> LEFT JOIN Purchase pu ON s.supplierID = pu.supplierID
-> GROUP BY s.supplierID, s.supplierName;
Query OK, 0 rows affected (0.036 sec)

MariaDB [SupermarketDB]> CREATE VIEW SalesSummary AS
-> SELECT DATE(s.saleDate) AS saleDate, COUNT(DISTINCT s.saleID) AS totalSales,
-> COALESCE(SUM(si.subTotal),0) AS totalRevenue,
-> COALESCE(SUM(s.pointsRedeemed), 0) AS totalPointsRedeemed
-> FROM Sale s
-> LEFT JOIN SaleItem si ON s.saleID = si.saleID
-> GROUP BY DATE(s.saleDate);
Query OK, 0 rows affected (0.038 sec)

MariaDB [SupermarketDB]> CREATE VIEW ProductsPerCategory AS

```

Views:

Figure A10. Customer Purchase History View

Shows the results generated by the CustomerPurchaseHistory view, providing customer, sale, product, quantity, pricing, discount, and subtotal information.

```
MariaDB [supermarketdb]> SELECT * FROM CustomerPurchaseHistory;
```

customerID	customerName	saleID	saleDate	productName	quantity	unitPrice	discount	subTotal
CUS001	John Mensah	SAL001	2026-08-01 00:00:00	Oats 500g	2	26.00	0.00	52.00
CUS001	John Mensah	SAL001	2026-08-01 00:00:00	Penne Pasta 500g	5	17.00	10.00	76.50
CUS001	John Mensah	SAL001	2026-08-01 00:00:00	A4 Printing Paper (Ream)	9	45.00	5.00	384.75
CUS002	Ama Boateng	SAL002	2026-08-01 00:00:00	Chocolate Bar	3	8.00	5.00	22.80
CUS003	Kwame Asante	SAL003	2026-08-01 00:00:00	Sprite 500ml	9	8.00	10.00	64.80
CUS003	Kwame Asante	SAL003	2026-08-01 00:00:00	Bathing Soap	6	9.00	0.00	54.00
CUS004	Akosua Owusu	SAL004	2026-08-01 00:00:00	Penne Pasta 500g	2	17.00	0.00	34.00
CUS004	Akosua Owusu	SAL004	2026-08-01 00:00:00	Palm Oil 1L	8	30.00	0.00	240.00
CUS005	Daniel Kofi	SAL005	2026-08-02 00:00:00	Bathing Soap	5	9.00	15.00	38.25
CUS005	Daniel Kofi	SAL005	2026-08-02 00:00:00	Notebook	1	12.00	15.00	10.20
CUS006	Abena Osei	SAL006	2026-08-02 00:00:00	Spaghetti 500g	8	16.00	0.00	128.00
CUS006	Abena Osei	SAL006	2026-08-02 00:00:00	Black Tea	7	14.00	15.00	83.70

Figure A11. Active Discounts View

Shows currently active discounts together with their discount details, applicable dates, products, and discount values.

```
MariaDB [supermarketdb]> SELECT * FROM ActiveDiscounts;
```

discountID	discountName	discountType	discountPercent	fixedAmount	startDate	endDate	productID	productName
DIS008	Back To School	Percentage	15.00	0.00	2026-08-01	2026-09-30	PRO008	Butter 250g
DIS011	Loyalty Discount	Percentage	5.00	0.00	2026-01-01	2026-12-31	PRO011	Sugar Bread
DIS012	Bulk Purchase	Percentage	10.00	0.00	2026-01-01	2026-12-31	PRO012	Chocolate Cake Slice

3 rows in set (0.069 sec)

Figure A12. Supplier Purchase Summary View

Shows a summary of purchase information for each supplier, including the number of purchases and the total purchase cost.

```
MariaDB [supermarketdb]> SELECT * FROM SupplierPurchaseSummary;
```

supplierID	supplierName	totalPurchases	totalPurchaseCost
SUP001	Accra Foods Ltd	1	2698.00
SUP002	Golden Harvest Ltd	1	1954.00
SUP003	Fresh Choice Ghana	1	3346.00
SUP004	Kumasi Wholesale	1	14362.00
SUP005	Prime Beverages	1	2468.20
SUP006	Healthy Dairy Ltd	1	9661.00
SUP007	Best Bakery Supply	1	5470.00
SUP008	Royal Canned Foods	1	8106.50
SUP009	Snack World Ltd	1	1116.00
SUP010	Morning Foods Ltd	1	12185.00
SUP011	Golden Grain Ltd	1	15692.00
SUP012	Pure Oils Ghana	1	15911.00
SUP013	Clean Home Ltd	1	8913.00
SUP014	Stationery Hub Ltd	1	11558.00
SUP015	Sweet Treats Ltd	1	3244.50
SUP016	Pure Water Ghana	1	818.00

16 rows in set (0.098 sec)

Figure A13. Sales Summary View

Shows daily sales information, including total sales, total revenue, and loyalty points redeemed.

```
MariaDB [supermarketdb]> SELECT * FROM SalesSummary;
```

saleDate	totalSales	totalRevenue	totalPointsRedeemed
2026-08-01	4	928.85	70
2026-08-02	4	585.60	0
2026-08-03	4	514.00	50
2026-08-04	4	565.00	70
2026-08-05	4	737.40	20
2026-08-06	4	386.10	40
2026-08-07	4	1278.85	60
2026-08-08	2	256.90	0

8 rows in set (0.026 sec)

• Stored procedures/functions

Procedures:

Figure A14. RecordPurchase Stored Procedure

Shows the creation of the RecordPurchase stored procedure for recording purchase transactions and their associated purchase items.


```

MySQL Client (MariaDB 12.2 (x64)) - "C:\Program Files\MariaDB 12.2\bin\mysql.exe" --defaults-file=C:\Program Files\MariaDB 12.2\data\my.ini -u root -p
MariaDB [SupermarketDB]>
MariaDB [SupermarketDB]> DELIMITER ;
MariaDB [SupermarketDB]> DELIMITER //
MariaDB [SupermarketDB]> CREATE PROCEDURE RecordPurchase(
-> IN p_purchaseID VARCHAR(10), IN p_supplierID VARCHAR(10), IN p_employeeID VARCHAR(10),
-> IN p_productID VARCHAR(10), IN p_quantity INT, IN p_unitCost DECIMAL(10,2))
-> BEGIN
-> DECLARE v_total DECIMAL(12,2);
-> INSERT INTO Purchase(purchaseID,supplierID,employeeID,purchaseDate,totalCost,status)
-> VALUES(p_purchaseID,p_supplierID,p_employeeID,CURDATE(),p_quantity*p_unitCost,'Completed');
-> INSERT INTO PurchaseItem(purchaseID,productID,quantity,unitCost) VALUES(p_purchaseID,p_productID,p_quantity,p_unitCost);
-> SELECT SUM(quantity*unitCost) INTO v_total FROM PurchaseItem WHERE purchaseID=p_purchaseID;
-> UPDATE Purchase SET totalCost=v_total WHERE purchaseID=p_purchaseID;
-> END //
Query OK, 0 rows affected (0.030 sec)

```

Figure A15. SearchProducts Stored Procedure

Shows the creation of the SearchProducts stored procedure for searching products by name, barcode, or category.

```

MariaDB [SupermarketDB]> SHOW CREATE PROCEDURE SearchProducts;
+-----+-----+-----+-----+
| Procedure | sql_mode | Create Procedure |
+-----+-----+-----+-----+
| SearchProducts | STRICT_TRANS_TABLES,ERROR_FOR_DIVISION_BY_ZERO,NO_AUTO_CREATE_USER,NO_ENGINE_SUBSTITUTION | CREATE DEFINER='root'@'localhost' PROCEDURE `SearchProducts`(IN p_search VARCHAR(100))
BEGIN
SELECT p.productID, p.productName, p.barcode, c.categoryName, p.unitPrice, i.quantity, p.status
FROM Product p
JOIN Category c ON p.categoryID = c.categoryID
JOIN Inventory i ON p.productID = i.productID
WHERE p.productName LIKE CONCAT('%', p_search, '%')
OR p.barcode = p_search
OR c.categoryName LIKE CONCAT('%', p_search, '%');
END | utf8mb4 | utf8mb4_uca1400_ai_ci | utf8mb4_uca1400_ai_ci |
+-----+-----+-----+-----+
1 row in set (0.100 sec)

```

Figure A16 CalculateSaleTotal Function

Shows the creation of the CalculateSaleTotal function used to calculate the total amount of a sale.

```

MariaDB [SupermarketDB]> SHOW CREATE FUNCTION CalculateSaleTotal;
+-----+-----+-----+-----+-----+-----+-----+-----+
| Function | sql_mode | Create Function |
+-----+-----+-----+-----+-----+-----+-----+
| CalculateSaleTotal | STRICT_TRANS_TABLES,ERROR_FOR_DIVISION_BY_ZERO,NO_AUTO_CREATE_USER,NO_ENGINE_SUBSTITUTION | CREATE DEFINER='root'@'localhost' FUNCTION 'CalculateSaleTotal'(p_saleID VARCHAR(10)) RETURNS decimal(12,2)
  READS SQL DATA
  DETERMINISTIC
BEGIN
  DECLARE v_total DECIMAL(12,2);
  SELECT COALESCE(SUM(subTotal),0) INTO v_total FROM SaleItem WHERE saleID = p_saleID;
  RETURN v_total;
END | utf8mb4 | utf8mb4_uca1400_ai_ci | utf8mb4_uca1400_ai_ci |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.027 sec)

```

• Trigger

Figure A17. AfterSaleItemInsert Trigger

Shows the AfterSaleItemInsert trigger, which automatically updates inventory quantity and the last-updated timestamp when a sale item is inserted.

```

DELIMITER ' at line 1
->
-> CREATE TRIGGER AfterSaleItemInsert
-> AFTER INSERT ON SaleItem
-> FOR EACH ROW
-> BEGIN
->   UPDATE Inventory SET quantity = quantity - NEW.quantity, lastUpdated = CURRENT_TIMESTAMP
->   WHERE productID = NEW.productID;
MariaDB [supermarketdb]> END //
->
-> DELIMITER ;

```